

Programming Fundamentals LAB – F25

Lab 07 – 21-10-2025

Turn on a LAB computer or your LAPTOP, Do Logon, and Enter Command Line Interface (CLI), the CMD window and observe the prompt. In drive D: or some other suitable drive create a folder pflab06 and change folder to that so that the prompt should be like **D:\pflab06>**.

Make sure that command g++ is recognized at the computer you have, also check the Notepad++.

To create or edit abc.cpp, use command: **D:\pflab06>start notepad++ abc.txt**

To compile abc.cpp, use command: **D:\pflab06>g++ abc.cpp -o abc.exe**

To run binary of abc.cpp, the abc.exe, use command: **D:\pflab06>abc.exe**

From the concepts learned in lectures so far, write the programs to solve the following problems.

Task 01 (20 marks for each)

1. Write a function that returns the sum of digits of its only int type parameter. Write another function **wrestlerInteger** that returns **true** value if sum of digits of its first int type parameter is more than sum of digits of its second int type parameter, otherwise it returns **false**. Later, call the function **wrestlerInteger** in main to demonstrate its working, saying x is a wrestler as it defeats y. Run the program several times for different integers and prepare a table with integer in first column and its divisors in second column. Also, test the sum of digits function by calling it in main. Can you guess about any flaw or flaws in the **wrestlerInteger** function? Please specify them too in comments.
2. In main store the whole statement of task 01, 1 in a string variable. Make a function and call in the declaration of above mentioned string variable to return the length (count of characters) in the first sentence (ends at .) in the string. Output the length followed by sequence of character to mimic as first sentence is displayed after its length. Later, to output the sentence, make a function with several necessary parameters, obviously the first one is the string that is containing the sentences. Demonstrate the working of this function.
3. Write a simple function to input and return an array of 11 doubles, a simple void function to output an array of 11 doubles, and main to test these functions. Your main task is to write a function to shift down the values in the array of 11 doubles. Shifting down means the values at i^{th} location of array should move to $(i+1)^{\text{th}}$ location of array. The last value of the array will

eventually be lost, and the original first value in the array should remain at its location. Now in main, create an array and input distinct 11 values in it, output the array, appropriately call rotation function, again output array. Now, use for loop to call rotation function 7 times, and then output the array. Notice values at the output now.

4. Write a function with two parameters, an int to search, and an array of 10 ints from which the other parameter is being searched, the function returns true or false accordingly. Test this function through some simple code in main. Your main task is to confirm that all the values in another array of 5 ints are present in the array of 10 ints mentioned earlier. Demonstrate the working of this function.
5. Read the guidelines provided after the thanks message below and write/demonstrate code to replace all the digits (leave other characters as they are) in a string with character #. A sample string to test is [my marks in 1st PF assignment are 88% in CLO 2.]

Thank you for your patience

The <cctype> header in C++ provides a set of useful functions for classifying and testing characters. They typically take a single parameter (the character to test) and return a non-zero value (which evaluates to true) if the character meets the condition, or zero (which evaluates to false) otherwise.

Function	Description	Example	Result
isalpha(c)	Checks if c is an alphabetic letter (A-Z or a-z).	isalpha('a')	True
isdigit(c)	Checks if c is a decimal digit (0-9).	isdigit('7')	True
isalnum(c)	Checks if c is alphanumeric (a letter or a digit).	isalnum('5')	True
islower(c)	Checks if c is a lowercase letter (a-z).	islower('A')	False
isupper(c)	Checks if c is an uppercase letter (A-Z).	isupper('A')	True
isspace(c)	Checks if c is a white-space character (space, tab, newline, etc.).	isspace(' ')	True
ispunct(c)	Checks if c is a punctuation character (any printable character that's not a space or alphanumeric).	ispunct('?')	True
isxdigit(c)	Checks if c is a hexadecimal digit (0-9, a-f, or A-F).	isxdigit('F')	True

The following functions change the case of a letter. If the character is not a letter of the appropriate case, it returns the character unchanged.

tolower(c) converts to its **lowercase** equivalent. And **toupper(c)** converts to its **uppercase** equivalent.

```
string text = "HeLlO";
cout << text[0] << " " << islower(text[0]) << endl;
cout << text[1] << " " << islower(text[1]) << endl;
cout << text[2] << " " << islower(text[2]) << endl;
cout << text[3] << " " << islower(text[3]) << endl;
cout << text[4] << " " << islower(text[4]) << endl;

int j;
for (j=0; j < 5; j=j+1)
{
    text[j] = tolower(text[j]); // c is converted to its lowercase equivalent
}

cout << "Lowercase: " << text << endl; // Output: hello
```